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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/037,592	05/18/2023	Jonathan Michael BLACKBURN	16742S-000003-US-NP	5069

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EXAMINER	
COUNTS, GARY W	

ART UNIT	PAPER NUMBER
1678	

NOTIFICATION DATE	DELIVERY MODE
08/27/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

troymailroom@hdp.com

Office Action Summary	Application No. 18/037,592	Applicant(s) BLACKBURN et al.	
	Examiner GARY W COUNTS	Art Unit 1678	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 07/22/26.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-9,11-12,20,22 and 27-31 is/are pending in the application.

5a) Of the above claim(s) 20 and 22 is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-9,11-12 and 27-31 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☑ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☐ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Status of the claims

The amendment filed 07/22/26 is acknowledged and has been entered. Claim 1 has been amended. New claims 27-31 have been added. Claims 21 and 23-26 have been canceled. Claims 10 and 13-19 were previously canceled. Claims 20 and 22 remain withdrawn as being directed to non-elected inventions. Accordingly, claims 1-9, 11-12 and 27-31 are under examination.

Claim Rejections - 35 USC § 101

35 U.S.C. 101 reads as follows:

> Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

Claims 1-9, 11-12 and 27-31 are rejected under 35 U.S.C. 101 because the claimed invention is directed to abstract ideas and/or to laws of nature/natural phenomena without significantly more.

The U.S. Patent and Trademark Office recently revised the MPEP with regard to § 101 (see the MPEP at 2106). Regarding the MPEP at 2106, in determining what concept the claim is "directed to," we first look to whether the claim recites:

(1) any judicial exceptions, including certain groupings of abstract ideas (*i.e.*, mathematical concepts, certain methods of organizing human activity such as a fundamental economic practice, or mental processes); and

(2) additional elements that integrate the judicial exception into a practical application (see MPEP § 2106.05(a)-(c), (e)-(h)).

Only if a claim (1) recites a judicial exception and (2) does not integrate that exception into a practical application, do we then look to whether the claim contains an "'inventive concept' sufficient to 'transform'" the claimed judicial exception into a patent-eligible application of the judicial exception. *Alice*, 573 U.S. at 221 (quoting *Mayo*, 566 U.S. at 82). In so doing, we thus consider whether the claim:

(3) adds a specific limitation beyond the judicial exception that is not "well-understood, routine, conventional" in the field (see MPEP § 2106.05(d)); or

(4) simply appends well-understood, routine, conventional activities previously known to the industry, specified at a high level of generality, to the judicial exception.
See MPEP 2106.

ELIGIBILITY STEP 2A: WHETHER A CLAIM IS DIRECTED TO A JUDICIAL EXCEPTION

Step 2A, Prong 1

The claims are directed to a naturally occurring correlation between autoantibodies in a serum or plasma sample of a rheumatoid arthritis patient prior to treatment with adalimumab and the patient having a good response or a poor response.

Step 2A, Prong 2

The additional elements of testing the sample for autoantibodies to SSB, TROVE2 and ZHX2 by use of an antigen array wherein the antigens are biotinylated

and each antigen being formed from a biotin carboxyl carrier protein does not apply, rely on, or use the judicial exception in a manner that imposes a meaningful limit on the judicial exception.

Also, with respect to the recitation “making a determination whether the patient will develop a good response or a bad poor response to treatment with adalimumab, based on the detection of said autoantibody biomarkers; wherein said autoantibody biomarkers comprise autoantibodies to antigens SSB, TROVE2 and ZHX2, wherein ZHX2 is associated with the good response, and SSB and TROVE2 are associated with the poor response”. The “determining” statement at best articulates the judicial exception, amounting only to a general instruction to apply or use the judicial exception. This could read on mental activity being performed solely in a practitioner’ head, e.g. A mental appreciation of the recited autoantibodies being correlated a good response or a poor response to adalimumab treatment in the subject. No active method steps are invoked or clearly required; the “determining” statement does not include any activity that would constitute a practical application, i.e. steps that apply, rely on or use the natural principle in a manner such that the claims amount to significantly more that the natural principal itself.

ELIGIBILITY STEP 2B: WHETHER THE ADDITIONAL ELEMENTS CONTRIBUTE AN "INVENTIVE CONCEPT"

Further, the additional elements of the claims are recited with a high level of generality and do not apply, rely on, or use the judicial exception in a manner that imposes a meaningful limit on the judicial exception. (the active method

steps/limitations recited in addition to the judicial exceptions themselves) and do not add significantly more to the judicial exception(s).

As shown by ImmuneTM (submitted in the IDS filed 08/29/23) it is well known, routine and conventional in the art to test a sample for autoantibodies to SSB, TROVE2 and ZHX2 and also additional autoantibody biomarkers (e.g. pages 1, 42, 46, 48). ImmneTM also shows that it is well known, routine and conventional to provide an antigen array comprising a panel of biotin carboxyl carrier protein.

With respect to the recitation "administering adalimumab to the patient if the determination is that the patient will develop the good response" as recited in instant claim 1. Although the claim invokes administering adalimumab to the patient, the claim currently recites "if the determination is that the patient will develop the good response". The recitation of "if" allows for the scenario of the determination of a bad response in the patient and thus allows for an embodiment wherein no adalimumab is administered. Therefore, this scenario does not recite something significantly more than the judicial exception.

It does not appear to be the case that the active steps recited, which are performed in order to gather the data or perform the assay, are steps recited or performed in an unconventional or non-routine way, such to provide an inventive concept under step 2B.

The claimed limitations as currently presented fail to recite limitations that add a feature that is more than well understood, conventional or routine in the field of diagnostics and biochemical assay methodologies.

For all of these reasons, the claims fail to include additional elements that are sufficient to either integrate the judicial exception(s) into practical application(s) thereof, or amount to significantly more than the judicial exception(s).

Response to Arguments

Applicant's arguments filed 07/22/26 have been fully considered but they are not persuasive.

101 Rejection:

Applicant argues that Claim 1 has been amended to further comprise "administering adalimumab to the patient if the determination is that the patient will develop the good response." Applicant submits that this reflects an active step that does not rely on or use the alleged natural principle.

This argument is not found persuasive because although the claim invokes administering adalimumab to the patient, the claim currently recites “if the determination is that the patient will develop the good response”. The recitation of “if” allows for the scenario of the determination of a bad response in the patient and thus allows for an embodiment wherein no adalimumab is administered. Therefore, this scenario does not recite something significantly more than the judicial exception. The claim does not positively determine the patient will have a good response and then administer the adalimumab to the patient determined to have a good response.

Applicant argues that new claims 27-31 recites the action of "providing an antigen array comprising a panel of biotinylated antigens bound to one or more streptavidin-coated substrates, each biotinylated antigen being formed from a Biotin

Carboxyl Carrier Protein folding marker which is fused in-frame with an antigen; wherein the antigens comprise SSB, TROVE2, and ZHX2."

This argument is not found persuasive because these limitations are considered to be additional elements and do not apply, rely on, or use the judicial exception in a manner that imposes a meaningful limit on the judicial exception. (the active method steps/limitations recited in addition to the judicial exceptions themselves) and do not add significantly more to the judicial exception(s).

Conclusion

No claims are allowed.

Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any nonprovisional extension fee (37 CFR 1.17(a)) pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of this final action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to GARY W COUNTS whose telephone number is (571)272-0817. The examiner can normally be reached M-F 7:00-4:00.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Gregory Emch can be reached at 571-272-8149. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/GARY COUNTS/
Primary Examiner, Art Unit 1678